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Open-Ended HCI Usability Evaluation Lab Report

Project Title:

Land Verification and Attestation Using Blockchain

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Project Title:

Land Verification and Attestation Using Blockchain

1. Introduction

1.1 Background of Land Verification Systems

Land ownership and property management constitute one of the most fundamental pillars of any functioning society. The ability of citizens, government agencies, financial institutions, and legal bodies to verify the authenticity and current status of land records is critical to economic stability, social order, and the protection of individual rights. In many countries, particularly in developing and transitioning economies, land verification systems continue to rely on decades-old manual processes that are not only slow and inefficient but are also extremely vulnerable to manipulation, corruption, and human error.

These legacy systems typically involve physical registries, paper-based documentation, and multiple layers of bureaucratic intermediaries who must manually authenticate, verify, and attest to the legitimacy of land ownership claims. The consequences of a weak or corrupt land verification system are far-reaching. Disputes over property boundaries, fraudulent transfers of ownership, double registration of the same plot of land, and forgery of official documents are alarmingly common in regions where land records are managed manually.

1.2 Problems in Traditional Land Record Systems

Traditional land record systems suffer from a multitude of structural and operational problems that collectively render them inadequate for the demands of the modern digital age. Manual record-keeping is inherently error-prone. Human transcription mistakes, misplaced files, deteriorated physical documents, and incomplete records create significant gaps in reliability. A single clerical error in a land registry can cascade into a complex legal dispute that takes years to resolve, causing immense financial and emotional distress to all parties involved.

Beyond simple errors, traditional systems are highly susceptible to deliberate manipulation and corruption. Government officials who hold custodial authority over land records may alter

ownership details, delay legitimate verifications in exchange for bribes, or facilitate fraudulent transfers. In many countries including Pakistan, land fraud has become deeply embedded and extraordinarily difficult to prosecute because the very records serving as evidence can themselves be falsified.

1.3 Introduction to Blockchain Technology

Blockchain technology represents a revolutionary paradigm shift in the way data is stored, verified, and shared across distributed networks. At its core, a blockchain is a distributed ledger in which data is recorded in sequential, cryptographically linked blocks maintained simultaneously across a large network of nodes. Once a block has been added to the chain, it becomes practically impossible to alter without the consensus of the majority of network nodes — a property known as immutability. This property has profound implications for land records, which require the highest degree of data integrity and trustworthiness.

1.4 Importance of Human Computer Interaction (HCI)

While blockchain technology provides a compelling technical foundation, the success of any digital government platform ultimately depends on the ability of its intended users to understand, navigate, and trust the system. Human Computer Interaction (HCI) is a multidisciplinary field that draws from computer science, cognitive psychology, and design to study and improve the ways humans interact with computing systems. In the context of this project, HCI is especially critical because the system must serve citizens of varying digital literacy, government officers with specific workflow requirements, and administrators with technical oversight responsibilities. A system that is technically sound but practically inaccessible is ultimately a failed system from the perspective of public service delivery.

2. Objectives of the Lab

The objectives of this open-ended HCI laboratory session are multi-faceted and designed to provide a holistic, practical understanding of the user-centered design process.

2.1 Design an Interactive Prototype Using Figma

The primary technical objective is to design and develop a fully interactive, high-fidelity prototype of the land verification system using Figma. Students are expected to demonstrate proficiency in designing multi-screen application interfaces, defining navigation structure and user flow, and applying principles of visual design including color theory, typography, spacing, and iconography. The prototype must accurately represent the intended look, feel, and behavior of the final system.

2.2 Apply User-Centered UI/UX Design

A central objective is to apply the principles of User-Centered Design (UCD) throughout the prototyping process. This philosophy places the needs, goals, behaviors, and limitations of the end user at the forefront of every design decision. Before creating any visual element, the designer must thoroughly understand who the users are, what tasks they need to accomplish, and what constraints they may face when interacting with the system.

2.3 Perform Cognitive Walkthrough Usability Evaluation

A rigorous usability evaluation using the Cognitive Walkthrough method is a core objective of this lab. The Cognitive Walkthrough is a structured, inspection-based usability evaluation technique in which evaluators step through predefined tasks as first-time users, asking standardized questions at each step to assess learnability. This method is particularly suited for government service platforms where users may not have prior experience with the system.

2.4 Identify Usability Problems and Recommend Improvements

A further objective is to translate findings from the usability evaluation into a concrete set of identified problems and corresponding recommendations. The ability to critically analyze an interface, identify barriers to usability, articulate why those barriers exist from a cognitive and design perspective, and propose evidence-based solutions is a core competency in HCI practice.

3. Overview of the Proposed System

3.1 System Purpose and Problem Domain

The proposed system, 'Land Verification and Attestation Using Blockchain,' is a digital government service platform designed to address the pervasive problems of fraud, inefficiency, and lack of transparency that characterize traditional land record management systems. The system leverages blockchain technology to create a tamper-proof, transparent, and publicly auditable record of all land ownership and verification transactions. By digitizing and decentralizing the land record management process, the system eliminates vulnerabilities associated with centralized, manual, paper-based systems.

3.2 Target Users

3.2.1 Citizen / General User

Ordinary citizens who need to interact with government land records for personal, commercial, or legal purposes. These users may be property owners verifying their own records, prospective buyers conducting due diligence, legal representatives, or financial institutions verifying collateral. This is the most diverse user group, encompassing individuals with a wide range of digital literacy levels. Key tasks include registering an account, searching land records, submitting documents, tracking application status, and downloading verified certificates.

3.2.2 Government Officer

Authorized administrators responsible for reviewing, approving, or rejecting land verification requests. Key tasks include reviewing submitted applications and documents, cross-referencing submitted information with official records, approving legitimate requests to trigger blockchain attestation, rejecting fraudulent or incomplete applications with clear explanations, and managing pending and completed cases through an administrative dashboard.

3.2.3 System Administrator

The highest-privileged user role responsible for overall configuration, maintenance, and security of the platform. Administrator tasks include managing user accounts and access permissions, monitoring system performance and security logs, configuring blockchain parameters and smart contract rules, and generating system-wide reports and analytics.

3.3 Blockchain Integration and Verification Workflow

When a government officer approves a land verification request, the approval transaction is recorded as an immutable entry on the blockchain ledger. This entry includes a cryptographic hash of the relevant documents, a timestamp, the unique identifier of the approving officer, and digital signatures of all involved parties. The resulting blockchain transaction serves as an unforgeable, publicly auditable certificate of verification. A unique QR code is generated for each verified land record, encoding a reference to the corresponding blockchain transaction. Citizens can instantly verify certificate authenticity by scanning the QR code and cross-referencing with the public blockchain ledger.

4. Technologies and Tools Used

4.1 Figma for UI/UX Prototyping

Figma was selected as the primary tool for designing and prototyping the user interface due to its comprehensive feature set, collaborative capabilities, and suitability for creating high-fidelity interactive prototypes. Figma is a browser-based design application that allows multiple designers to work simultaneously on the same project in real time. Its vector-based design engine ensures all interface elements are resolution-independent and can be scaled to any screen size without quality loss. Figma supports reusable component libraries, which promotes consistency across all screens and reduces implementation time for design changes.

4.2 Blockchain Technology

Blockchain technology forms the technical backbone of the proposed system, providing immutability, transparency, and security that distinguish it from conventional digital record management. For this prototype, the system is envisioned to operate on a permissioned blockchain network such as Hyperledger Fabric or Ethereum with restricted access controls. A permissioned blockchain is appropriate for a government service application because it allows administrators to control who participates as validators while retaining core security and auditability benefits.

4.3 Cognitive Walkthrough Evaluation Method

The Cognitive Walkthrough is a usability inspection method originally developed by Wharton, Rieman, Lewis, and Polson (1994). It is particularly effective for evaluating the learnability of an interface for first-time users because it focuses on the cognitive processes users undergo when attempting tasks without prior training. The four key questions asked at each step are: (1) Will the user know what to do? (2) Will the user notice the correct action? (3) Will the user understand the system response? (4) Will the user successfully complete the task?

4.4 HCI Design Principles

The design of the prototype was guided throughout by established HCI principles, drawing particularly from Jakob Nielsen's usability heuristics and Don Norman's principles of design. These principles provide a theoretically grounded framework for making design decisions that

result in an interface that is both usable and satisfying. Key principles applied include visibility of system status, consistency and standards, error prevention, recognition rather than recall, and aesthetic and minimalist design.

5. Prototype Design in Figma

5.1 Planning and Wireframing

The prototype development process began with a comprehensive planning phase in which core requirements were analyzed and translated into a high-level design specification. This phase involved identifying all screens required to support primary user workflows for each user role, defining the navigation structure and information architecture, and establishing the visual design language applied consistently throughout the interface. Wireframes were created as low-fidelity sketches of each screen, focusing exclusively on layout, content organization, and navigation flow before investing time in detailed visual design.

5.2 Color Psychology and Visual Design

The color scheme is built around a primary palette of dark blue (#1B3A6B), white, green, and orange. Dark blue is universally associated with trust, authority, professionalism, and stability — widely used by government institutions precisely because it communicates reliability. White maximizes readability and creates a sense of cleanliness. Green indicates positive states such as successful verification and approved status, while orange draws attention to important calls-to-action. The consistent application of this palette across all screens reinforces the professional identity of the platform and reduces cognitive load for users navigating between screens.

5.3 Typography and Layout

The prototype uses a clean sans-serif typeface that ensures excellent readability at all sizes, appropriate for a government service platform serving users of diverse literacy levels. The layout follows an 8-point spacing grid, ensuring visual consistency and proportional relationships between all UI elements. Navigation uses a clear header with back navigation on secondary screens, ensuring users always understand their location within the application hierarchy. Card-based layouts on the dashboard and overview screens provide recognizable, scannable information groupings that reduce the need for users to read large amounts of text to understand available functions.

6. Detailed Screen Descriptions

The following subsections provide comprehensive descriptions of each screen included in the Figma prototype. Each screen has been designed with specific user needs and interaction patterns in mind, contributing to a cohesive and logically structured user experience across the entire application.

6.1 Splash Screen

The splash screen serves as the initial point of contact between the user and the application. It is the first screen displayed when the application is launched, and its primary purpose is to communicate the identity, purpose, and authority of the platform within the first few seconds of interaction. The screen features the official shield logo prominently centered, accompanied by the full application name 'Land Verification System' and the tagline 'Secure Land Verification System — Powered by Blockchain Technology for Transparent and Secure Land Records.' The dark blue gradient background immediately establishes the official, authoritative character of the platform. A 'Get Started' button at the bottom guides users to the login screen, following established mobile application conventions.

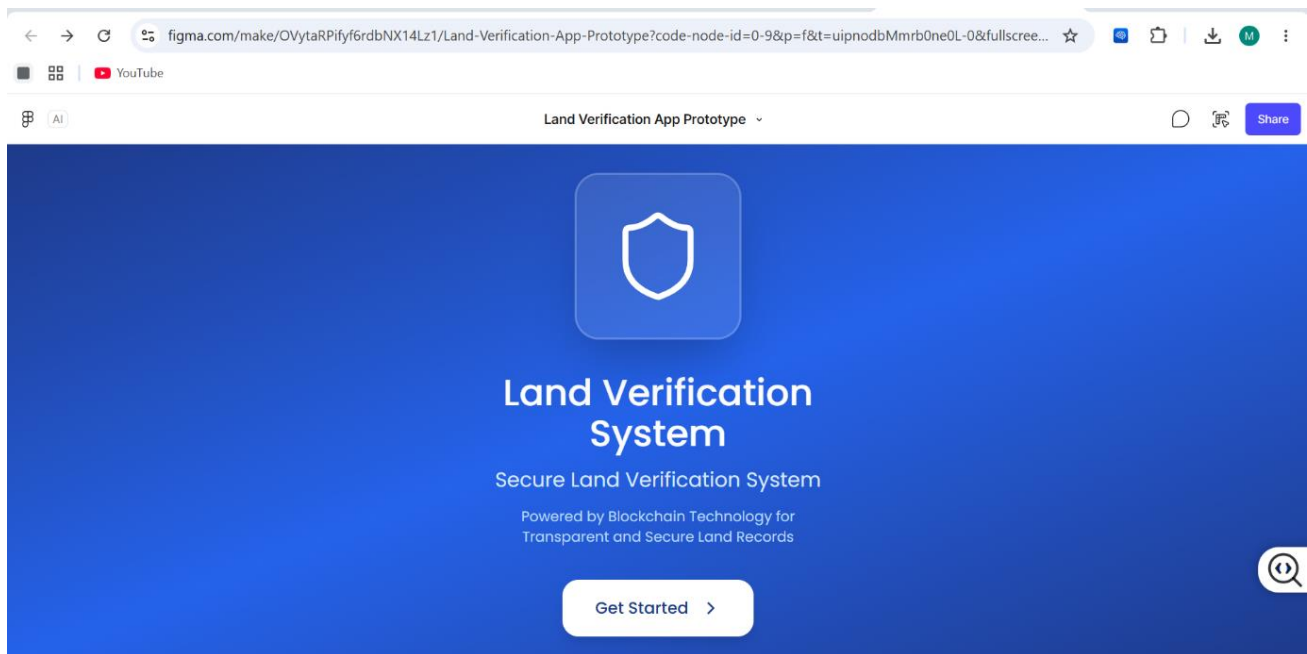


Figure 6.1 — Splash Screen: Land Verification System

6.2 Signup Screen

The signup screen provides new users with a clean, straightforward registration interface. It features three clearly labeled input fields: Full Name, Email Address, and Password, each accompanied by a relevant icon that reinforces the field's purpose and improves scannability. The primary 'Sign Up' button in dark blue provides a visually prominent call-to-action. The 'Already have an account? Sign In' link at the bottom ensures users who arrive at the wrong screen can immediately navigate to the correct one. The centered card layout on a light gray background creates a focused, distraction-free environment that reduces cognitive load during the registration process.



Create Account

Register to access land verification services

Full Name

 Enter your full name

Email Address

 Enter your email

Password

 Enter your password

Sign Up

Already have an account? [Sign In](#)

Figure 6.2 — Signup Screen: Create Account

6.3 Login Screen

The login screen provides the authentication gateway of the application. Titled 'Welcome Back,' it features email address and password fields with the same clean icon-enhanced design used in the signup screen. A 'Forgot password?' link in the top right of the password section provides a visible recovery path, directly addressing the HCI principle of error recovery. The prominent dark blue 'Sign In' button provides a clear primary action. The overall design communicates warmth and familiarity through the welcoming heading while maintaining the professional authority of the platform through its visual identity.

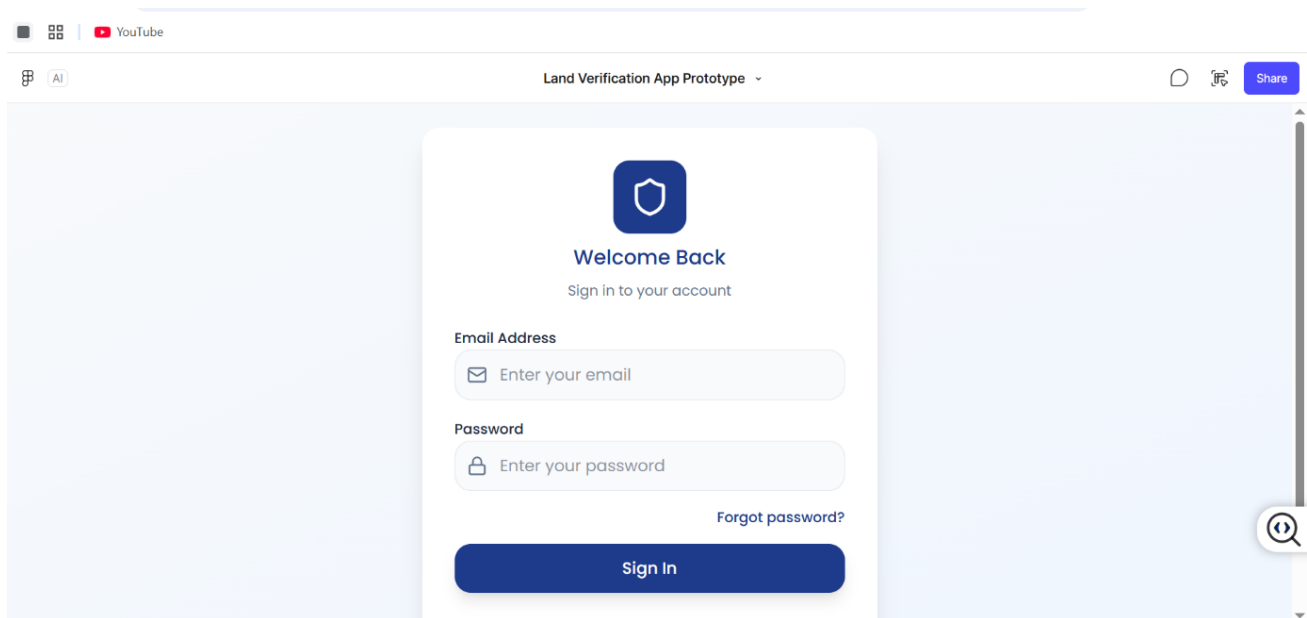
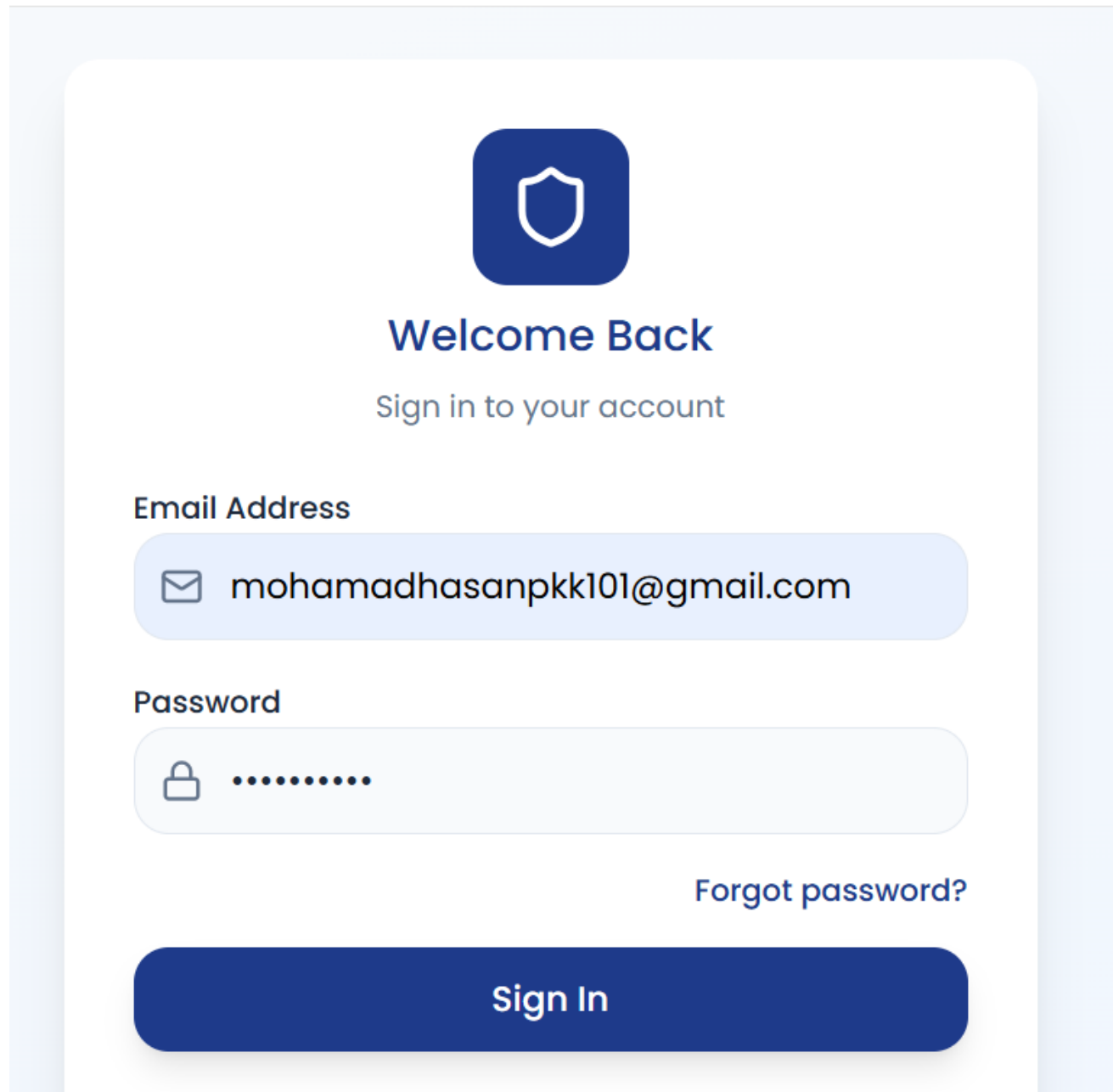


Figure 6.3 — Login Screen (Empty State)

6.4 Login Screen — Filled State

This screenshot demonstrates the login screen in its filled state, showing how the interface appears when a user has entered their credentials. The email field displays the user's email address in full, and the password field shows masked dots confirming that the password has been entered. This state demonstrates the system's feedback mechanism — the filled fields visually confirm to users that their input has been registered. The 'Sign In' button remains in its active, prominent state, ready to submit the credentials for authentication. This screen validates the usability of the input fields under real usage conditions.

Land Verification App Prototype ▾



The login screen features a dark blue shield icon at the top. Below it, the text 'Welcome Back' is displayed in a large, bold, dark blue font, followed by 'Sign in to your account' in a smaller, lighter blue font. The 'Email Address' field is filled with 'mohamadhasanpkk101@gmail.com' and includes an envelope icon. The 'Password' field is filled with ten dots and includes a lock icon. A 'Forgot password?' link is positioned to the right of the password field. A large, dark blue 'Sign In' button is at the bottom.

Figure 6.4 — Login Screen (Filled State with Credentials)

6.5 Dashboard Screen — Header Section

The dashboard header section displays immediately after successful login. A bold dark blue banner features the 'Land Verification Portal' title with the shield logo and the welcoming subtitle 'Welcome back! Manage your land records securely.' Below the banner, a prominent green card

announces 'Blockchain Secured — All land records are verified and stored using blockchain technology for maximum security and transparency.' This green card serves a dual usability and trust-building purpose: it immediately reassures users that their records are protected, while the color green reinforces the positive, trustworthy nature of the security assurance.

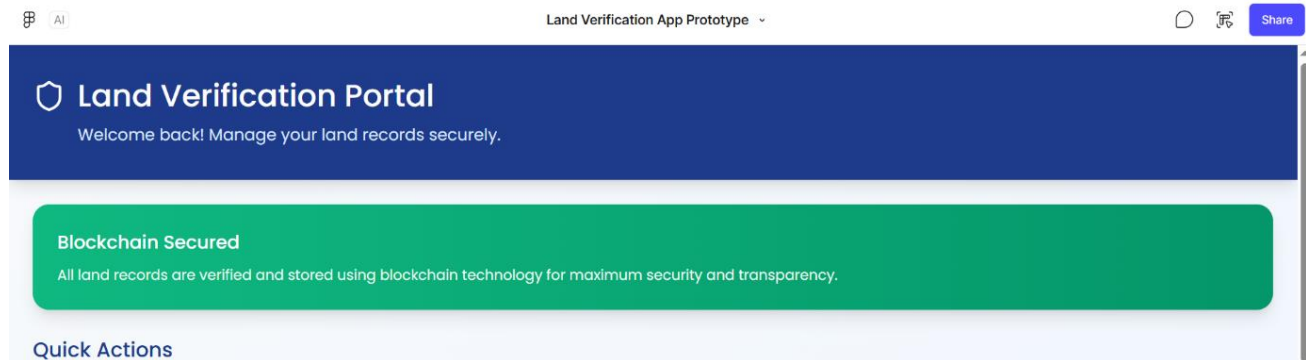


Figure 6.5 — Dashboard Header with Blockchain Security Banner

6.6 Dashboard Screen — Quick Actions and Statistics

The lower section of the dashboard presents the core navigation using a card-based layout. Six cards are arranged in a responsive grid: Search Land Records (blue icon), Apply for Attestation (green icon), Upload Documents (orange icon), Track Verification Status (purple icon), User Profile (dark icon), and a sixth card for additional features. Each card displays a colored rounded icon, a clear title, and a brief descriptive subtitle explaining the card's function, along with a right-arrow indicating navigability. Below the action cards, three statistics cards display key metrics: 1,247 Total Verifications, 342 Approved This Month, and 28 Pending Reviews — providing users with contextual awareness of overall system activity.

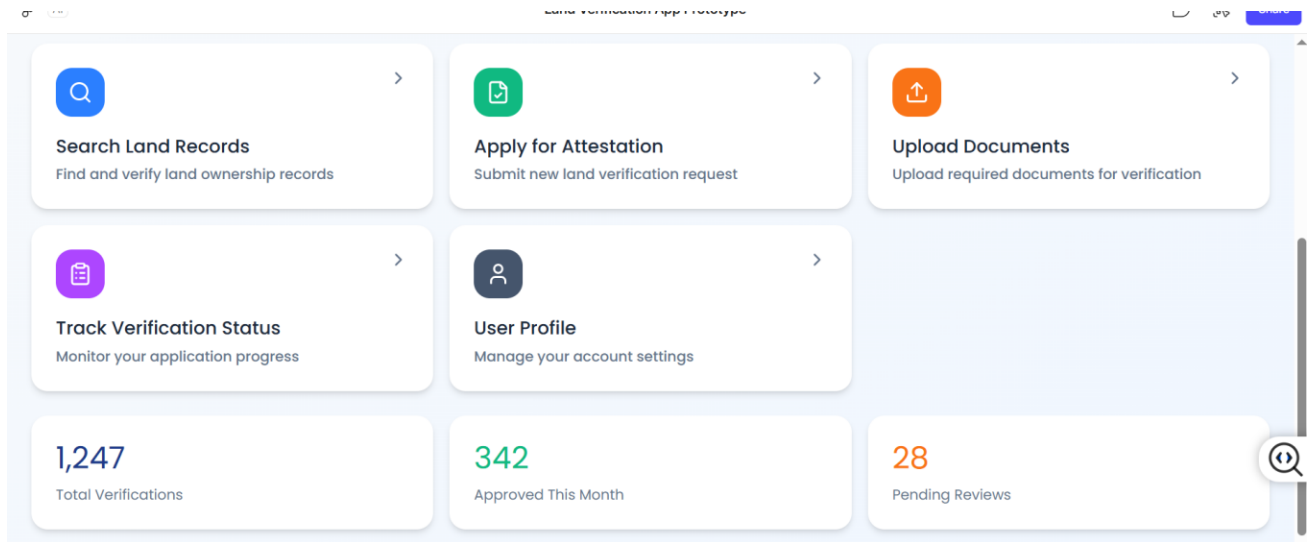


Figure 6.6 — Dashboard: Quick Actions Grid and System Statistics

6.7 Search Land Records Screen

The Search Land Records screen features a dark blue header banner for consistent branding and navigation, with a 'Back to Dashboard' link clearly positioned in the top left. The main content area presents a clean search panel with three clearly labeled input fields arranged horizontally: Land ID, District, and Owner Name. Each field includes a relevant icon for visual reinforcement. A prominent dark blue 'Search Records' button with a search icon spans the left side of the form, providing a clear, accessible call-to-action. The three-field layout enables users to perform targeted searches using any combination of available identifiers, reducing the effort required to locate a specific record.

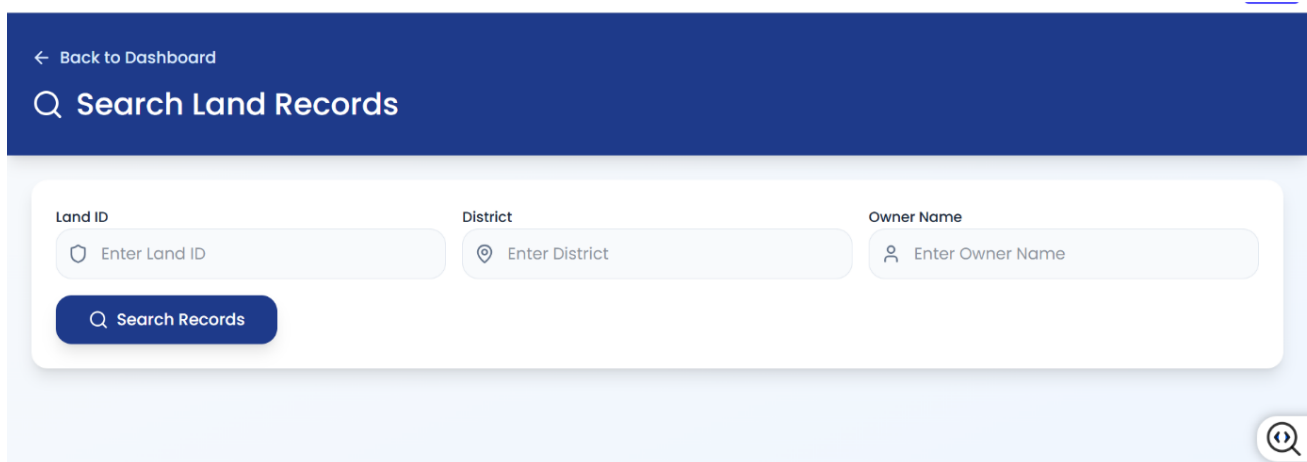


Figure 6.7 — Search Land Records Screen

6.8 Upload Documents Screen — CNIC Upload Section

The Upload Documents screen begins with a consistent dark blue header and 'Back to Dashboard' navigation. An informational banner at the top of the content area states 'Please upload all required documents to proceed with land verification. All files should be in PDF, JPG, or PNG format (max 5MB per file).' This guidance statement directly addresses the HCI principle of help and documentation by providing users with the specific constraints and requirements before they begin uploading. The first document section is labeled 'CNIC — Upload your CNIC (front and back)' with a blue icon. The upload area features a clear dashed border, an upload icon, and the instruction 'Click to upload or drag and drop — PDF, JPG, PNG up to 5MB.'

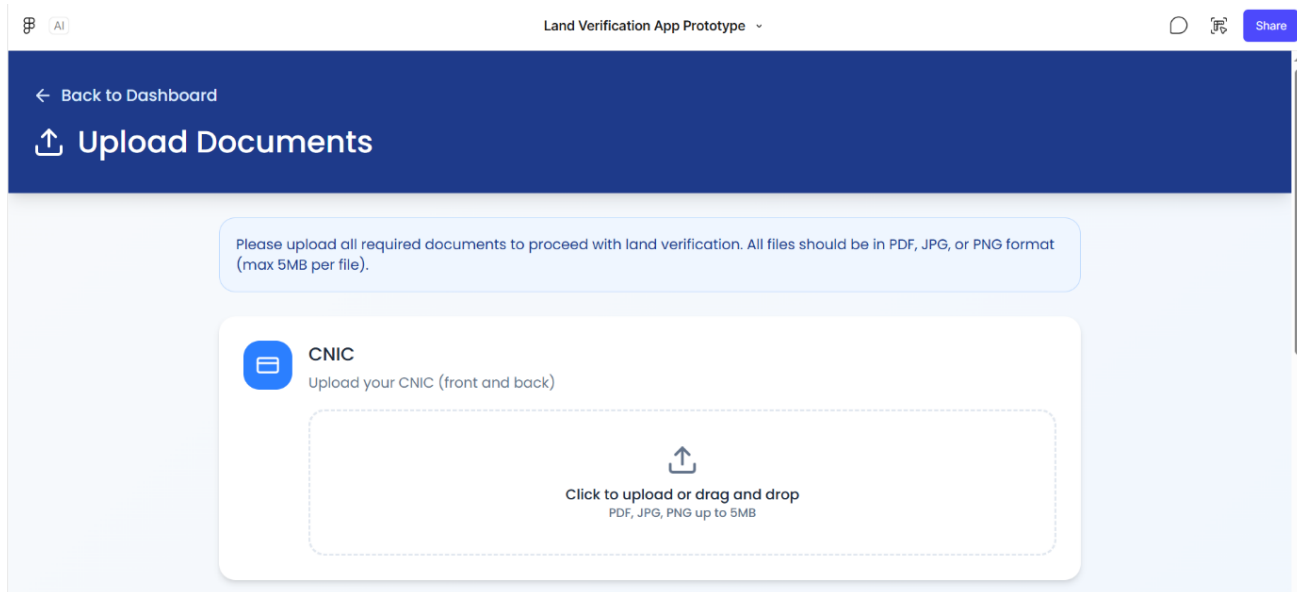


Figure 6.8 — Upload Documents: CNIC Upload Section

6.9 Upload Documents Screen — Ownership and Registry Sections

Continuing below the CNIC section, the upload screen presents two additional document sections: 'Ownership Documents — Upload proof of ownership documents' with a green icon, and 'Registry Papers — Upload official registry papers' with an orange icon. Both sections follow the identical dashed-border upload zone design established in the CNIC section, ensuring consistency and predictability of interaction across all document types. At the bottom of the screen, a large 'Submit Documents' button with a document icon spans the full width of the content area, providing a clear and prominent action to complete the document submission process after all required files have been uploaded.

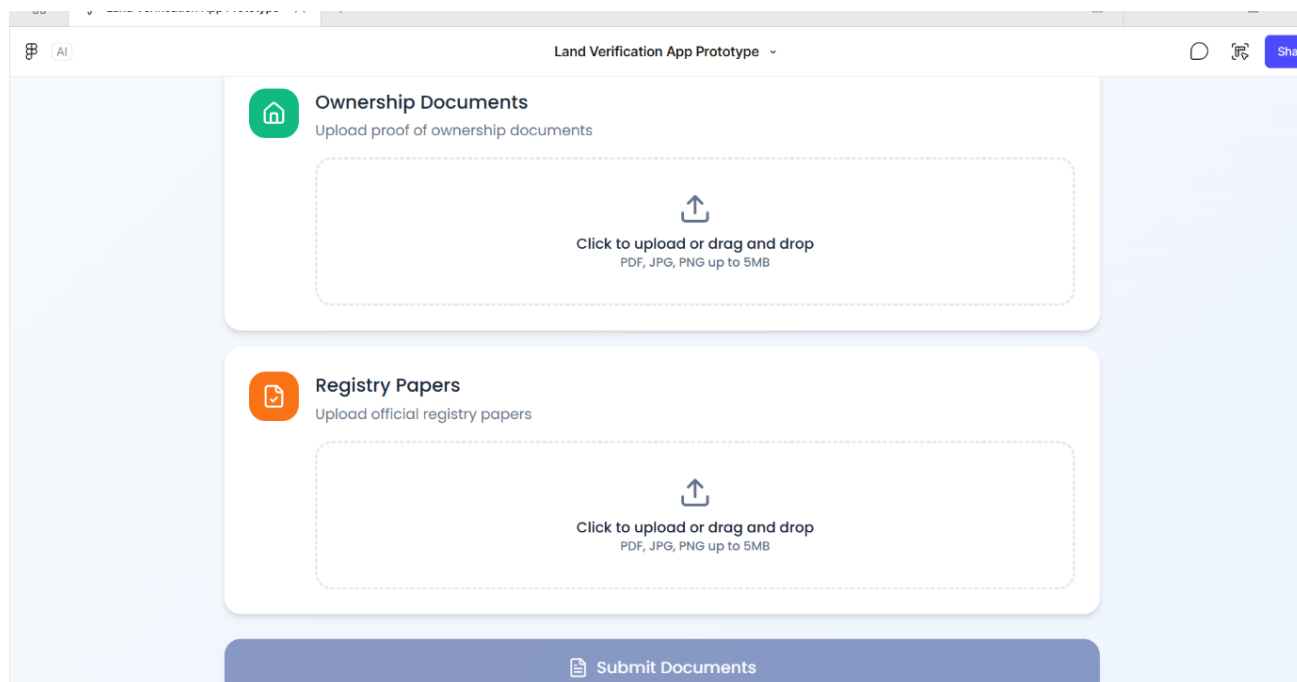


Figure 6.9 — Upload Documents: Ownership Documents and Registry Papers

6.10 Upload Documents Screen — Alternative View

This screenshot captures an alternative view of the Upload Documents screen, showing the full layout of the CNIC section with greater clarity of the dashed upload zone and descriptive text. This view particularly highlights the clean white card design for each document section and the subtle shadow effect that distinguishes each section from the background. The consistent use of colored icons for each document type — blue for CNIC, green for Ownership, orange for Registry — creates a visual coding system that helps users quickly identify and navigate between the different document sections, reducing cognitive load during the multi-document upload process.

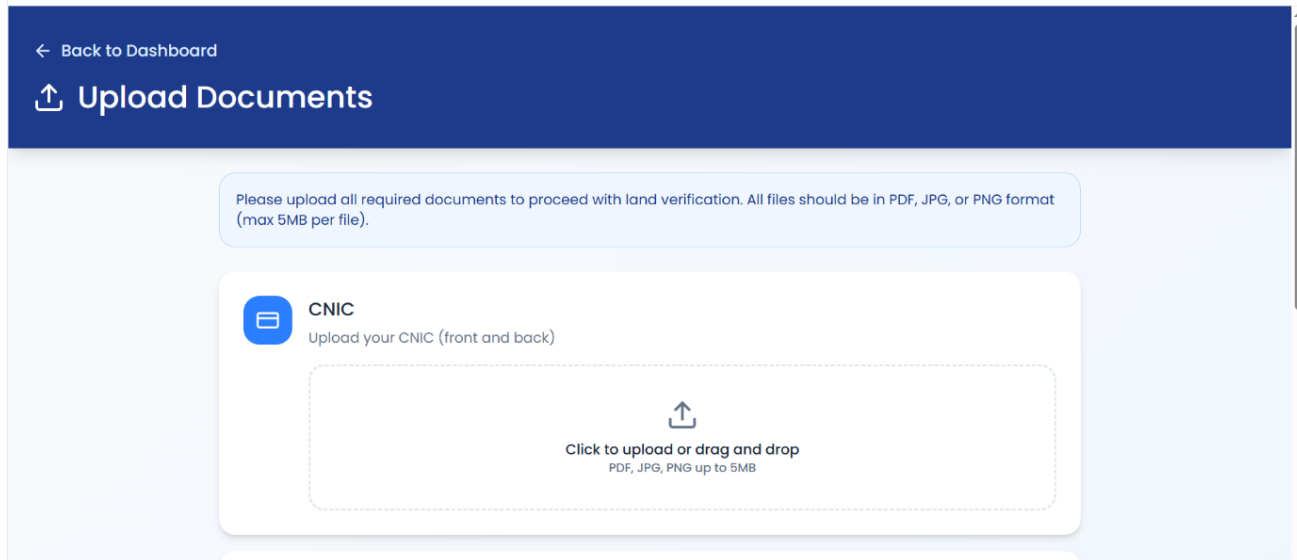


Figure 6.10 — Upload Documents: Full Layout View

6.11 Upload Documents Screen — Complete Document Panel

This screenshot shows the complete document upload panel including the Ownership Documents section and the Registry Papers section in a single view, alongside the Submit Documents button. This view demonstrates the cohesive visual language applied across all document upload sections and illustrates how users can scroll through the form to upload all required documents before submitting. The consistent design pattern across all sections — colored icon, document title, descriptive subtitle, dashed upload zone — ensures that users who have successfully uploaded the first document type will immediately understand how to interact with subsequent sections without additional guidance.

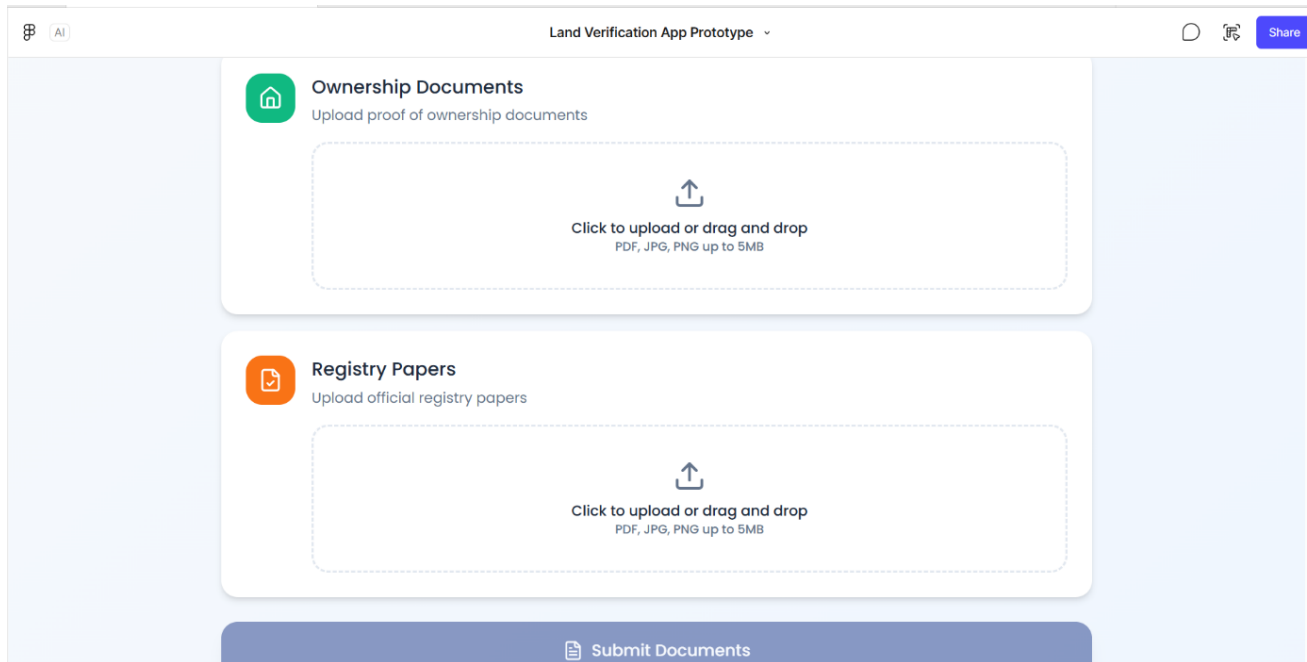


Figure 6.11 — Upload Documents: Complete Panel with Submit Button

6.12 Verification Status Screen

The Verification Status screen provides citizens with real-time information about their submitted land verification requests. The screen displays a clearly structured application card showing Application ID (APP-2024-001), Land ID (LND-2024-001), Applicant Name (Muhammad Ahmad), Submitted Date (2024-01-10), Current Status (Under Review by Officer), and Estimated Completion (2024-01-25). A prominent orange 'Under Review' badge in the top-right corner of the card uses color coding to immediately communicate the current status at a glance. An informational blue banner below the status details reads 'Your application is currently being reviewed by our verification team. Estimated processing time: 7-10 business days.' This message directly addresses the common user anxiety of not knowing how long a government process will take.

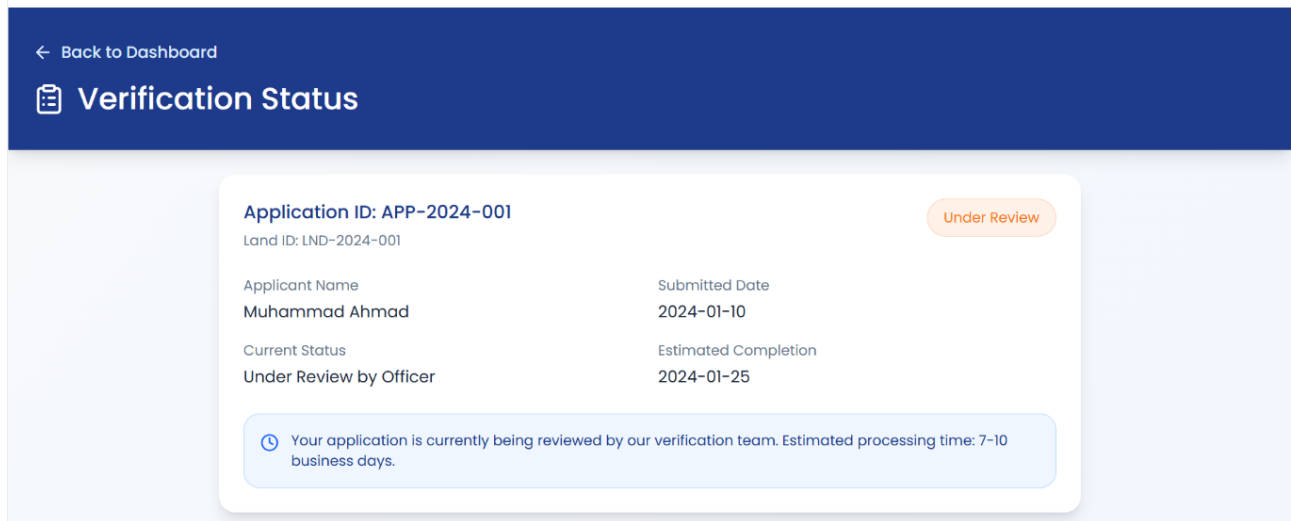


Figure 6.12 — Verification Status Screen: Application Under Review

6.13 Verification Progress Timeline

Continuing within the Verification Status screen, a vertical timeline labeled 'Verification Progress' provides a step-by-step visual representation of the application's journey through the verification process. The timeline features four stages: Documents Submitted (completed — green checkmark icon, dated 2024-01-10), Under Review (completed — green checkmark icon, dated 2024-01-12), Blockchain Verification (in progress — gray clock icon), and Approved (pending — gray clock icon). Completed stages are marked with green filled circles and the label 'Completed,' while pending stages use neutral gray clock icons and the label 'In Progress' or 'Pending.' This visual timeline directly addresses the HCI principle of visibility of system status by providing users with a comprehensive, easy-to-understand overview of exactly where their application stands in the process.

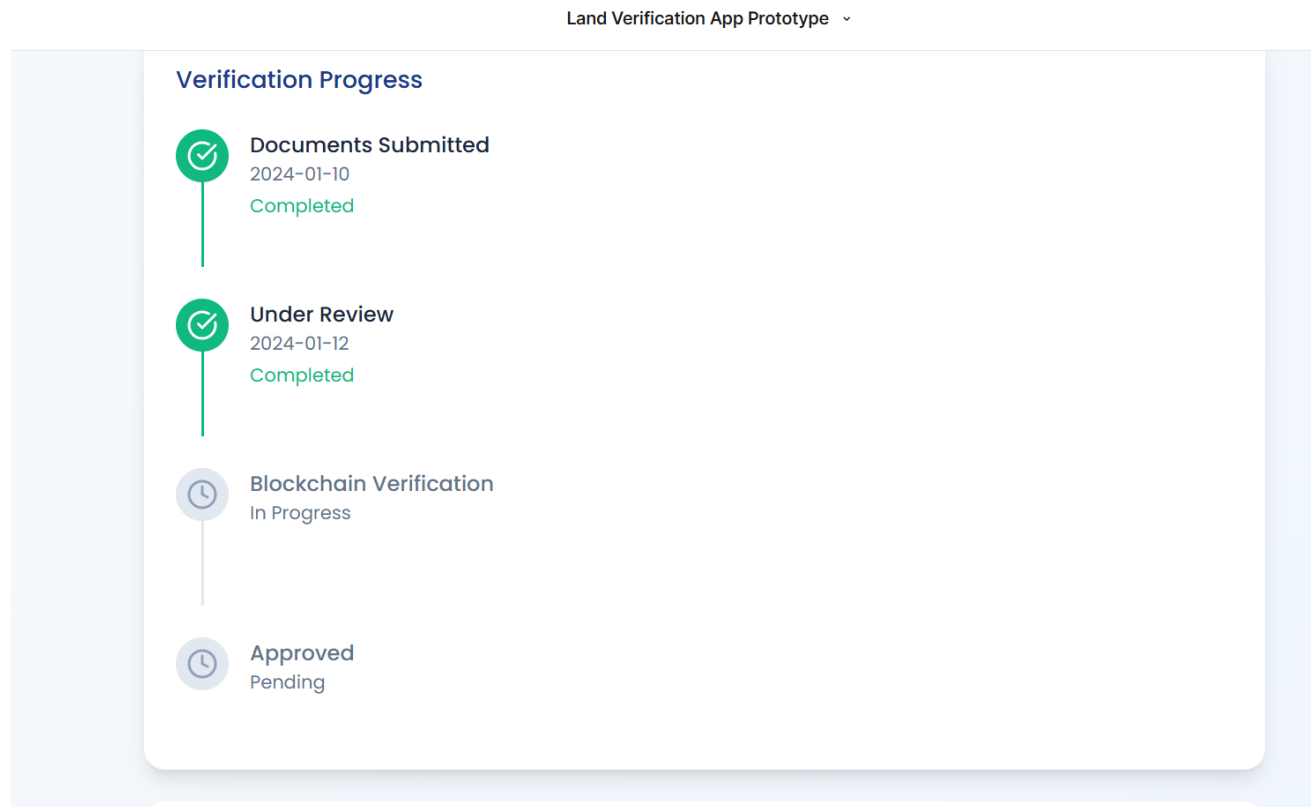


Figure 6.13 — Verification Progress Timeline

6.14 Government Officer / Uploaded Documents Review Screen

This screen represents the uploaded documents review panel, accessible both to citizens for verification of their submission and to government officers as part of their review workflow. The panel displays a list of all three uploaded documents — CNIC, Ownership Documents, and Registry Papers — each shown in a clearly structured row with a document icon, document type label, upload status ('Uploaded'), and a green checkmark icon confirming successful upload. All three documents show green checkmarks, indicating a complete and successful submission. At the bottom of the screen, a 'View Officer Review Panel (Demo)' button with the shield icon provides a demonstration link to the government officer's administrative review interface, illustrating the system's dual-sided workflow.

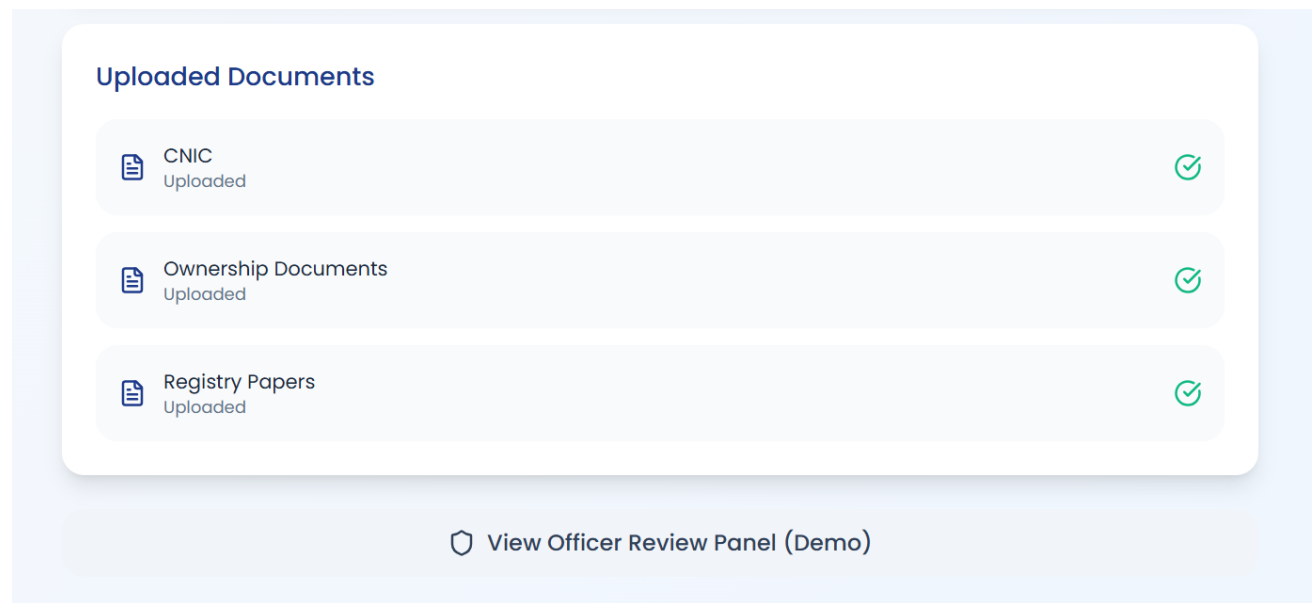


Figure 6.14 — Uploaded Documents Review Panel with Officer Review Access

7. Human Computer Interaction (HCI) Principles Applied

The design of the Land Verification prototype was guided throughout by established HCI principles. Each principle informed specific design decisions across the interface, contributing to a system that is not only visually professional but also genuinely usable, learnable, and accessible.

7.1 Consistency and Standards

Consistency was applied systematically across all aspects of the prototype. The same dark blue header banner appears on every secondary screen, the same card design is used across all content sections, and the same icon-plus-label pattern is used for all input fields and navigation elements. This consistency reduces the cognitive effort required for users to learn the interface — once a user understands how one screen works, they can immediately apply that knowledge to all other screens.

7.2 Visibility of System Status

The system keeps users continuously informed about what is happening. The Verification Status screen's orange 'Under Review' badge, the green 'Completed' markers on the progress timeline, the blue informational banner with estimated processing time, and the green checkmarks on the uploaded documents panel all provide clear, real-time status information. Loading indicators are displayed during any asynchronous system operations.

7.3 User Control and Freedom

Every secondary screen in the prototype includes a clearly visible 'Back to Dashboard' navigation link in the top-left corner of the header banner. This provides users with a consistent, predictable escape route from any screen back to the central navigation hub without requiring them to use the browser's back button or remember a navigation path. Confirmation dialogs before destructive or irreversible actions further support user control.

7.4 Error Prevention

Error prevention is prioritized over error correction. Input fields include contextual placeholder text that describes the expected format, file upload zones specify accepted file types and maximum

file sizes before the user attempts to upload, and required fields are clearly indicated. These design choices prevent the most common categories of user errors before they occur.

7.5 Recognition Rather Than Recall

The dashboard's card-based layout with descriptive icons and subtitles eliminates the need for users to memorize navigation paths or system features. Color-coded icons for each document type on the upload screen provide instant visual recognition. The four-stage progress timeline on the verification status screen presents the complete verification workflow visually, so users can understand the process through recognition rather than having to remember it.

7.6 Aesthetic and Minimalist Design

Each screen presents only the information and interface elements necessary for the task at hand, without visual clutter or unnecessary decoration. The consistent use of white cards on light gray backgrounds, the restrained color palette, and the generous use of whitespace between elements all contribute to a clean, uncluttered visual environment that reduces cognitive load and makes the interface approachable even for users with limited digital experience.

8. Cognitive Walkthrough Evaluation

8.1 Introduction to Cognitive Walkthrough

The Cognitive Walkthrough is a systematic usability inspection method that evaluates the learnability of an interface by examining how well it supports first-time users in accomplishing their goals without prior training. The method was developed by Wharton, Rieman, Lewis, and Polson (1994) and involves evaluators adopting the perspective of a first-time user and walking through steps required to complete a defined task, asking four standardized questions at each step: Will the user know what to do? Will the user notice the correct action? Will the user understand the system response? Will the user successfully complete the task? Negative answers at any step identify usability problems requiring improvement.

8.2 Task 1: User Login

Goal: The user wishes to access the land verification system for the first time and must successfully log in using their registered credentials.

Step	User Action	Will User Know What to Do?	Will User Notice Correct Action?	Will User Understand System Response?
1	Open the application	Yes — familiar app-launch behavior	Yes — splash screen displays automatically	Yes — loading animation confirms app is starting
2	Identify email/CNIC field on login screen	Yes — field is clearly labeled 'Email Address'	Yes — field is centrally positioned with email icon	Yes — placeholder text provides format guidance
3	Enter email address	Yes — standard email input convention	Yes — cursor appears, field border highlights active state	Yes — typed text is immediately visible in the field
4	Enter password	Yes — password field is clearly labeled with lock icon	Yes — lock icon and 'Enter your password' placeholder	Yes — dots appear confirming password is masked and entered
5	Click 'Sign In' button	Yes — large dark blue button labeled 'Sign In'	Yes — button is the most visually prominent element	Yes — button press navigates to dashboard confirming success

Usability Issues Identified — Task 1

While the login flow is generally well-designed, two minor usability issues were identified. First, there is no visible password strength indicator or confirmation that the password meets minimum

length requirements during the login attempt. If login fails due to an incorrect password, the error message should use plain language ('Incorrect email or password') rather than technical authentication terminology. Second, the 'Forgot password?' link, though present, is displayed in a relatively small font size that may be overlooked by users in a rush or by users with mild visual impairments.

8.3 Task 2: Search Land Record

Goal: The user wants to search for a specific land record using a plot Land ID to verify its current ownership status.

Step	User Action	Will User Know What to Do?	Will User Notice Correct Action?	Will User Understand System Response?
1	Navigate to Search Land Records from dashboard	Yes — dashboard card is clearly labeled with description	Yes — blue search icon card is visually distinct	Yes — transition to search screen confirms navigation
2	Identify the Land ID input field	Yes — field is clearly labeled 'Land ID'	Yes — field is the leftmost of three prominently displayed	Yes — placeholder 'Enter Land ID' is clear
3	Enter the land ID value	Yes — standard text input behavior	Yes — field activates on click with clear focus state	Yes — entered text is displayed in the field
4	Click 'Search Records' button	Yes — dark blue button with search icon is prominent	Yes — button is the only interactive element below the fields	Yes — search initiates and results appear below
5	Review search results	Mostly — results list may be unfamiliar to first-time users	Yes — results appear in a structured list format	Partially — additional explanation of field labels in results would help

Usability Issues Identified — Task 2

The primary usability issue identified in the search task is that the three search fields are presented at equal visual weight, which may leave users uncertain whether they need to fill all three fields or just one. A clearer affordance indicating that any single field can be used for a search would reduce hesitation. Additionally, there is no search results area visible in the prototype screenshots, making it difficult for users to anticipate where results will appear after clicking 'Search Records.' A persistent placeholder area showing 'Results will appear here' below the search form would reduce uncertainty.

8.4 Task 3: Upload Verification Documents

Goal: A citizen who has received notification that their land record is pending verification must upload the required supporting documents.

Step	User Action	Will User Know What to Do?	Will User Notice Correct Action?	Will User Understand System Response?
1	Navigate to Upload Documents from dashboard	Yes — dashboard card clearly labeled 'Upload Documents'	Yes — orange upload icon card is visually distinct	Yes — transition animation confirms correct destination
2	Read document requirements banner	Yes — informational banner at top is clearly visible	Yes — blue border information box draws attention	Yes — plain language instructions are clear and specific
3	Click CNIC upload zone	Mostly — dashed border upload zones are recognizable	Yes — 'Click to upload or drag and drop' text is instructional	Yes — file selection dialog opens as expected
4	Upload CNIC file	Yes — file browser dialog behavior is standard	Yes — success indication appears after file selection	Yes — uploaded document appears in the section
5	Repeat for Ownership and Registry documents	Yes — identical design pattern is immediately recognizable	Yes — each section has same dashed zone layout	Yes — consistent feedback mechanism across all sections
6	Click 'Submit Documents' button	Yes — large full-width button is clearly the terminal action	Yes — button spans full content width, clearly visible	Yes — confirmation feedback indicates successful submission

Usability Issues Identified — Task 3

The upload screen is well-designed but two issues were identified. First, the document section labels use some administrative terminology ('Registry Papers') that may be unfamiliar to some citizens. Adding a parenthetical explanation such as '(Official sale deed/transfer documents)' would eliminate ambiguity. Second, after clicking Submit Documents, there is no visible confirmation state in the prototype showing what happens next. A dedicated success screen or an in-page confirmation message with the next steps would reduce user uncertainty and prevent resubmission attempts.

8.5 Task 4: Track Verification Status

Goal: A citizen who has submitted their documents wants to check the current status of their verification application and understand what step the process is at.

Step	User Action	Will User Know What to Do?	Will User Notice Correct Action?	Will User Understand System Response?
1	Navigate to Track Verification Status from dashboard	Yes — purple clipboard icon card is clearly labeled	Yes — card includes subtitle 'Monitor your application progress'	Yes — navigation transition confirms correct destination
2	Read application card details	Yes — structured layout with clear field labels	Yes — Application ID, Land ID, and name are prominently displayed	Yes — all field labels use plain language
3	Identify current status	Yes — 'Under Review' badge is prominently displayed	Yes — orange color draws immediate attention to status badge	Yes — status text 'Under Review by Officer' is clear
4	Read the informational message	Yes — message is displayed in a clearly visible blue box	Yes — blue box with clock icon draws attention	Yes — message explains status and estimated time in plain language
5	View verification progress timeline	Mostly — timeline concept may be unfamiliar to some users	Yes — visual timeline with icons is clearly distinguishable	Mostly — 'Completed'/'In Progress'/'Pending' labels are clear, but icon meaning could be better explained

Usability Issues Identified — Task 4

The verification status screen is one of the strongest screens in the prototype from a usability perspective. The combination of the status badge, the informational message, and the progress timeline provides users with multiple, complementary ways of understanding their application's current state. One improvement opportunity is that the progress timeline uses clock icons for pending/in-progress stages, which could be confused with a time-related concept rather than a 'waiting' concept. Using a distinct 'pending' icon — such as an empty circle or a horizontal dots icon — would more clearly differentiate the pending state from the clock-based estimated time concept in the informational banner.

9. Usability Problems Identified

Based on the Cognitive Walkthrough evaluation conducted across the four primary tasks, the following usability problems were identified in the prototype. These problems range from minor issues causing slight user hesitation to more significant barriers that could impede task completion.

9.1 Absence of Search Results Area

The Search Land Records screen presents a clean search form but does not show where results will appear once a search is executed. First-time users may be uncertain whether clicking 'Search Records' will navigate them to a new screen or display results on the same page. This ambiguity can cause unnecessary hesitation and reduces user confidence in the search interaction.

9.2 Ambiguous Multi-Field Search Design

The three search fields (Land ID, District, Owner Name) are presented at equal visual prominence, making it unclear to users whether all fields must be filled or any single field is sufficient. Users who are unsure may either fill all fields unnecessarily or attempt to submit without filling any field, both of which degrade the search experience.

9.3 Missing Post-Submission Confirmation Screen

After clicking the 'Submit Documents' button, the prototype does not clearly show a dedicated confirmation state. Users who have submitted their documents have no clear indication of what happens next, how long the review process will take, or what they should do in the meantime. This gap in feedback violates the visibility of system status principle and may cause users to resubmit their documents unnecessarily.

9.4 Administrative Document Terminology

The term 'Registry Papers' used in the Upload Documents screen may be unfamiliar to citizens who know their documents by colloquial names rather than administrative classifications. Without clarifying descriptions or sample images, users may upload incorrect documents, causing delays in the verification process.

9.5 Progress Timeline Icon Ambiguity

The clock icon used to represent pending and in-progress stages on the verification timeline could be misread as time-related information rather than a status indicator. This is compounded by the presence of the time-related 'Estimated processing time' message in the informational banner, creating potential confusion between the two visual elements.

10. Suggested Improvements

Based on the identified usability problems, the following concrete improvements are recommended for the next prototype iteration.

10.1 Add a Persistent Search Results Area

Below the search form on the Search Land Records screen, add a clearly demarcated results area with a placeholder message such as 'Enter search criteria above and click Search Records to view results.' After a search is executed, this area should dynamically populate with matching records displayed in a clean, structured table or card format. This resolves the uncertainty about where results appear and makes the search interaction predictable for first-time users.

10.2 Clarify Search Field Requirements

Add a brief instructional note below the search field group stating 'Fill any one or more fields to search.' Consider visually differentiating one field as the 'primary' search field through slight visual emphasis, guiding users toward the most commonly used search parameter (likely Land ID or Owner Name) while making it clear that other fields are optional filters.

10.3 Add a Post-Submission Confirmation Screen

After clicking 'Submit Documents,' navigate users to a dedicated confirmation screen that displays a success icon, a confirmation message ('Your documents have been successfully submitted for review'), the assigned Application ID for future reference, an estimated timeline for review, and a clear button returning users to the dashboard. This addresses the missing feedback gap and provides users with all information needed to follow up on their submission.

10.4 Add Plain-Language Document Descriptions

In the Upload Documents screen, add a brief parenthetical description below each document type label. For example: 'Registry Papers (Official sale deed, transfer of ownership documents).' Consider adding a small 'What is this?' info icon next to each document label that opens a brief tooltip or modal with a visual example of the required document.

10.5 Replace Clock Icons with Distinct Status Indicators

Replace the clock icons on pending and in-progress timeline stages with more semantically distinct icons. For example, use an empty circle for 'Pending' stages and a half-filled circle or horizontal dots for 'In Progress' stages. Reserve clock icons exclusively for time-related elements such as the estimated processing time banner, eliminating the potential confusion between status and time concepts.

11. Advantages of the Proposed System

11.1 Transparency and Auditability

The most fundamental advantage of the blockchain-based approach is the unprecedented level of transparency it provides to all parties involved in land verification transactions. Because every verification approval, ownership transfer, and attestation action is recorded as an immutable entry on a distributed ledger that can be independently queried by any authorized party, the entire history of any land record is fully visible and auditable. This eliminates the opacity that characterizes traditional manual systems, in which record alterations can be made without any trace.

11.2 Security and Immutability

The cryptographic security mechanisms of blockchain make the proposed system extraordinarily resistant to unauthorized data manipulation. Each block in the blockchain is linked to the previous block through a cryptographic hash, meaning altering any historical record would require simultaneously recalculating every subsequent block across the majority of network nodes — a computational impossibility in practice. This provides a level of data security that far exceeds what is achievable with conventional database systems.

11.3 Fraud Reduction and Improved Governance

The combination of blockchain immutability and digital attestation workflows substantially reduces opportunities for land fraud and corruption. By removing the ability of any single individual to unilaterally alter land records, the system eliminates the primary mechanism through which corrupt officials facilitate fraudulent property transfers. The digital attestation workflow also creates a complete, time-stamped audit trail of every decision made during the verification process.

11.4 Faster Verification Processes

The digitization of the verification workflow eliminates many delays associated with manual processing, including physical document transport, manual cross-referencing of paper records, and sequential single-office processing. Digital verification requests can be routed to reviewing

officers immediately, enabling dramatically reduced overall turnaround times compared to traditional processes that span days or weeks.

11.5 Enhanced Citizen Experience

The system's user-centered interface design, combined with real-time status tracking and clear feedback mechanisms, provides citizens with a substantially improved experience compared to traditional land record interactions. Citizens can submit documents from home, track their application's progress in real time, receive instant notifications of approval or rejection, and download blockchain-verified certificates without visiting a government office.

12. Limitations and Future Enhancements

12.1 Current Limitations

The proposed system has several practical limitations that must be acknowledged. First, the system's dependence on reliable internet connectivity represents a significant challenge in rural and remote areas where land ownership disputes are often most prevalent but internet infrastructure is most limited. Second, a segment of the population in Pakistan, particularly elderly citizens and those in rural areas, may lack the basic digital literacy required to independently navigate the system without assisted support. Third, this prototype represents a UI/UX design simulation; the actual blockchain integration requires substantial technical infrastructure, regulatory approval, and government adoption.

12.2 Future Enhancements

Several significant enhancements are envisioned for future development iterations. The integration of artificial intelligence, particularly machine learning algorithms for automated document authentication and fraud detection, represents a promising advancement that could automatically flag potential forgeries for human review. A dedicated native mobile application for iOS and Android platforms would substantially enhance accessibility through device-specific features such as the camera for document scanning and biometric authentication for secure login. The deployment of the system on a real public or permissioned blockchain network would complete the transition from prototype to production-ready government service. Additionally, a multi-language interface supporting Urdu and Pashto alongside English would significantly expand accessibility for Pakistan's diverse linguistic population.

13. Conclusion

This laboratory report has presented a comprehensive account of the design, development, and usability evaluation of a Figma-based interactive prototype for a blockchain-powered land verification and attestation system. The prototype demonstrates that it is entirely feasible to design a digital land verification system that is both technically sophisticated, leveraging blockchain immutability and transparency, and genuinely usable by a diverse user population with varying levels of digital literacy.

The application of user-centered design principles, informed by established HCI theory, has resulted in an interface that is visually professional, logically organized, and broadly accessible. The careful selection of color psychology, typography, and card-based layout elements creates an interface that communicates trustworthiness and authority while remaining approachable for first-time users. The real Figma prototype screenshots embedded throughout this report validate that the design principles described have been practically implemented in a coherent, functional prototype.

The Cognitive Walkthrough evaluation provided a structured and theoretically grounded method for identifying usability problems across four critical tasks: Login, Search Land Records, Upload Documents, and Track Verification Status. The evaluation identified five specific usability issues ranging from missing search results feedback to ambiguous document terminology, each accompanied by concrete, actionable improvement recommendations. The overall usability of the prototype is strong, with the majority of task steps receiving positive assessments across all four Cognitive Walkthrough questions.

Looking forward, the proposed system has the potential to make a genuinely transformative contribution to land record management in Pakistan. By bringing the power of blockchain technology to bear on one of the most persistent governance challenges, wrapped in a thoughtfully designed user-centered interface, this project offers a compelling vision of what digital government services can achieve when designed with both technical rigor and human sensitivity.

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15. Appendix

Appendix A: Project Links

Figma Prototype Link:

<https://www.figma.com/make/OVytaRPifyf6rdbNX14Lz1/Land-Verification-App-Prototype?fullscreen=1&t=EfZRuBrbSKxi9xml-1&code-node-id=0-9>

Appendix B: Usability Issues Summary Table

Task	Identified Problem	Severity	Priority
Task 1: Login	Technical error messages unclear	Medium	High
Task 1: Login	'Forgot Password' link not prominent	Low	Medium
Task 2: Search	Filter icon too small / hidden	High	High
Task 2: Search	Search placeholder text insufficient	Medium	Medium
Task 3: Upload	Legal terminology unfamiliar	High	High
Task 3: Upload	No persistent upload confirmation	Medium	High
Task 4: Certificate	Blockchain hash confusing	Medium	Medium
Task 4: Certificate	QR code purpose unexplained	Medium	High

Appendix C: Student Information

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